# AMC Grade 7-8 Preparation For Round 2

1. Which of the following is the greatest value?

A) $\sqrt{\sqrt[3]{5 \cdot 6}}$ B) $\sqrt{6\sqrt[3]{5}}$ C) $\sqrt{5\sqrt[3]{6}}$ D) $\sqrt[3]{6\sqrt{5}}$

$\sqrt[6]{30}$ $\sqrt{\sqrt[3]{6^3 \times 5}}$ $\sqrt{\sqrt[3]{5^2 \times 6}}$ $\sqrt[3]{\sqrt{6^2 \times 5}}$

$\sqrt[6]{1080}$ $\sqrt[6]{750}$ $\sqrt[6]{180}$

$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$

2. Four numbers, when added to three numbers combined at the same time result in 180, 197, 208, and 222. What is the largest four numbers?

A) 66 B) 96 C) 89 D) 52

Let the 4 numbers be a, b, c, d

When we add 3 of them together at a time we get 180, 197, 208, 222

$$\begin{cases} a+b+c = 180 \\ a+b+d = 197 \\ a+c+d = 208 \\ b+c+d = 222 \end{cases}$$

$+$

$3a+3b+3c+3d = 807$

$a+b+c+d = \frac{807}{3} = 269$

- $a = (a+b+c+d) - (b+c+d) = 269 - 222 = 47$
- $b = 269 - 208 = 61$
- $c = 269 - 197 = 72$
- $d = 269 - 180 = 89$

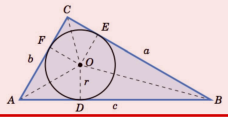
Thus, the largest of the 4 number is 89.

3. A circle inscribed in $\triangle ABC$ has side lengths 8 cm, 10 cm, and 12 cm, as shown in the figure. The length of side $BE$ is equal to:

A) 12 B) 5 C) 68 D) 4

Th: $\Delta ABC$ inscribed in a circle, let $s = \frac{1}{2}(a,b,c)$ (semi-perimeter)
the radius of the circle:

$$R = \frac{abc}{4\sqrt{s(s-a)(s-b)(s-c)}}$$




$2s = a+b+c$

$2s = (AD+BD)+(BE+CE)+(AF+FC)$

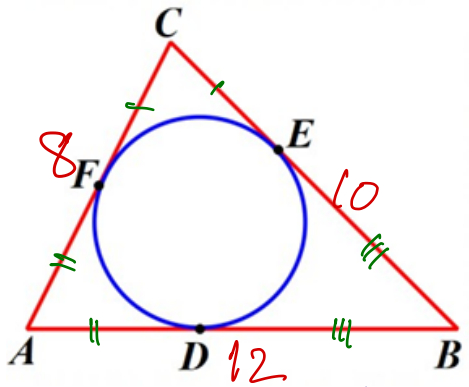
but $AD = AF$, $BE = BD$, $CE = CF$

$\Rightarrow 2s = 2AD + 2BD + 2CE$

$s = AD + \underbrace{BE+CE}_{a}$

$s = AD + a$

$\therefore \boxed{AD = s - a}$




$AD = s - a$

$AD = 15 - 10$

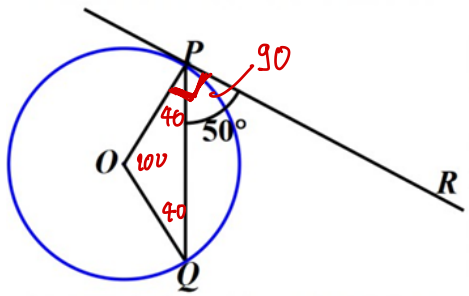
$\boxed{AD = 5}$

$s = \frac{8+10+12}{2} = 15$

$a = 10$

4. In the figure, if $O$ is the center of the circle, $PQ$ is the chord, and the line $PR$ touches the circle at $P$, forming an angle of 50° with the chord $PQ$, then $\angle POQ$ is equal to:

A) 100° B) 120° C) 112° D) 130°



5. For a natural number, when it increases by 12, it is equal to 160 times its multiplicative inverse, so that number is:

$\to n$ $\quad n+12 \quad = \quad 160 \times \frac{1}{n}$

A) 26 B) 8 C) 6 D) 32

$n+12 = 160 \times \frac{1}{n}$

$n(n+12) = 160$

$n^2+12n-160=0$

$\Delta = b^2-4ac = 12^2-4(1)(-160) = 784$

$\sqrt{\Delta} = \sqrt{784} = 28$

$n = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-12 \pm 28}{2}$ $\begin{cases} n=-20 \ \times \\ n=8 \ \checkmark \end{cases}$

natural numbers are always positive

or

$n^2-8n+20n-160=0$

$n(n-8)+20(n-8)=0$

$(n-8)(n+20)=0$

$n-8=0,\ n=8 \ \checkmark$

$n+20=0,\ n=-20 \ \times$

6. A takes 10 days less than B's time to complete a task. If A and B can work together to complete the task in 12 days, find the time B alone would take to finish the work.

A) 15 days B) 18 days C) 25 days D) 30 days

$\star$ Work rate $(R) = \frac{\text{Work}}{\text{time}} \Rightarrow \text{Work} = R \times t$

$t_A = t_B - 10$

$R_{A+B} = \frac{1}{12}$ $\to$ 100% of the task $\to$ 12 days to complete

$\Leftrightarrow R_A + R_B = \frac{1}{12}$

$\frac{1}{t_A} + \frac{1}{t_B} = \frac{1}{12}$

$\frac{1}{x} + \frac{1}{x-10} = \frac{1}{12}$

$12(x-10+x) = x(x-10)$

$24x-120 = x^2-10x$

$x^2-34x+120=0$

$(x-30)(x-4)=0$

$x=30,\ x=4 \ \times$

since $t_A$ would be $4-10=-6$

7. At a certain time of sunlight, a 60-meter-tall building casts a 42-meter shadow. How far is the shadow of a 10-meter-high traffic light at the same time?

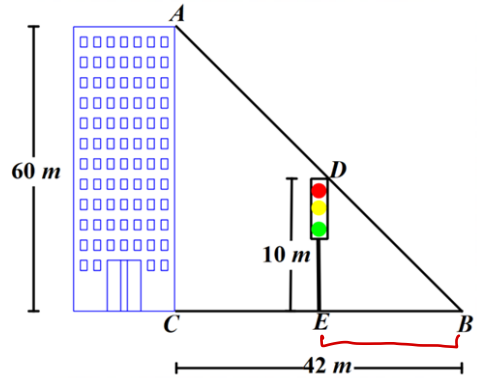
A) 6 m B) 12 m C) 7 m D) 5 m

$\Delta ABC \sim \Delta DBE$

$\frac{AC}{DE} = \frac{BC}{BE}$

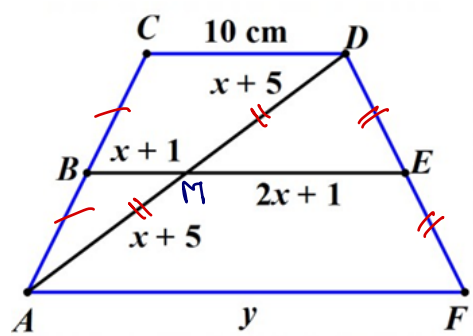
$\frac{60}{10} = \frac{42}{x}$

$x = \frac{\cancel{42}^{7} \times \cancel{10}}{\cancel{60}_{\cancel{10}}} = 7$



8. Determine the length of the diagonal $AD$ in the trapezoid below if $CD \parallel AF$, and $B$ and $E$ are the midpoints of $AC$ and $DF$, respectively.

A) 16 cm B) 8 cm C) 27 cm D) 18 cm




$\Delta ACD \sim \Delta ABM$

$\frac{CD}{BM} = \frac{AD}{AM}$

$\frac{10}{x+1} = \frac{2\cancel{(x+5)}}{\cancel{x+5}}$

$\frac{10}{x+1} = 2$

$10 = 2x+2$

$5 = x+1$

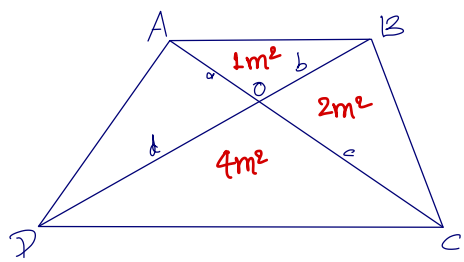
$x = 4$

$\Rightarrow AD = 2(4+5) = 2 \times 9 = \boxed{18}$

quadrilateral

9. Let ~~rectangle~~ $ABCD$ have diagonals $AC$ and $BD$ intersecting at $O$. If the areas of $\triangle AOB$, $\triangle BOC$, and $\triangle COD$ are equal to $1\,\text{m}^2$, $2\,\text{m}^2$, and $4\,\text{m}^2$ respectively, find the area of $\triangle AOD$ in $\text{m}^2$.

A) $2\,\text{m}^2$ B) $6\,\text{m}^2$ C) $8\,\text{m}^2$ D) $3\,\text{m}^2$




Method 1 : Guess

Draw the quadrilateral,

$A_{\Delta AOD} \simeq 2$ (similar to $\Delta_{\Delta BOC}$ by just looking)

$\Rightarrow$ option $2m^2$ & $3m^2$ are ok

but $3m^2$ would be too large

$\therefore$ $2m^2$ is the correct answer.

Method 2 : Do it the right way!

By proportional of Area of triangle in a quadrilateral :

$\frac{A_{\Delta AOB}}{ab} = \frac{A_{\Delta BOC}}{bc} = \frac{A_{\Delta COD}}{cd} = \frac{A_{\Delta AOD}}{ad} = k$

$\Leftrightarrow \begin{cases} A_{\Delta AOB} = abk = 1 & (1) \\ A_{\Delta BOC} = bck = 2 & (2) \\ A_{\Delta COD} = cdk = 4 & (3) \end{cases}$

where $k$ is constant

$(1) \div (2) : \frac{a}{c} = \frac{1}{2} \Rightarrow c = 2a \; (4)$

$(2) \div (3) : \frac{b}{d} = \frac{1}{2} \Rightarrow d = 2b \; (5)$

From (1) : $abk = 1 \Rightarrow b = \frac{1}{ak}$

sub into (5) : $d = \frac{2}{ak}$

$\therefore A_{\Delta AOD} = kad = k \times a \times \frac{2}{ak} = \boxed{2}$

10. In the trapezoid $ABCD$ below, find the area of the striped part.

A) 25 B) 36 C) 24 D) 16

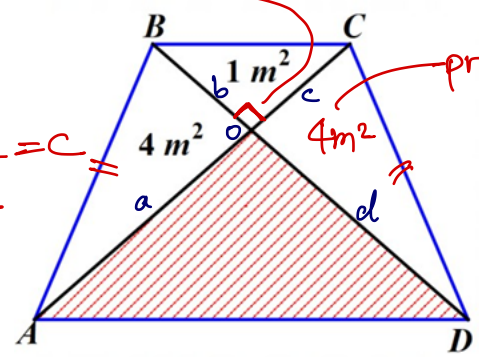
$$\frac{1}{bc} = \frac{4}{ab} = \frac{4}{cd} = \frac{x}{ad} = K$$

$$\begin{cases} \frac{1}{bc} = \frac{4}{ab} \Rightarrow a = 4c \\ \frac{1}{bc} = \frac{4}{cd} \Rightarrow d = 4b \\ \frac{4}{ab} = K \Rightarrow a = \frac{4}{Kb} \end{cases}$$

$$\Rightarrow x = Kad = K\left(\frac{4}{Kb}\right)(4b) = 16$$

$$\therefore \boxed{A_{\Delta AOD} = 16\,m^2}$$

Assuming




probably

$4m^2$

$$\frac{b^2}{2} = 1$$

$$b = \sqrt{2} = c$$

$$\frac{ab}{2} = 4$$

$$ab = 8$$

$$a = \frac{8}{\sqrt{2}}$$

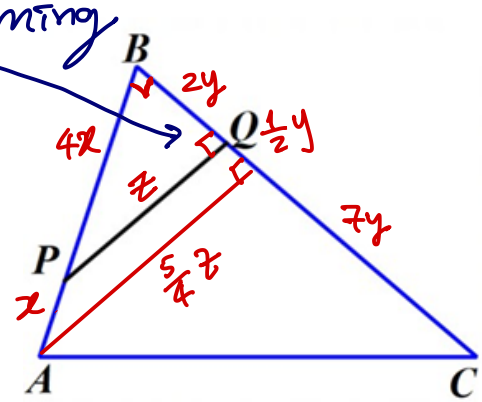
$$a = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

$$\Delta AOD = \frac{ad}{2} = \frac{a^2}{2} = \frac{(4\sqrt{2})^2}{2} = \frac{16 \times 2}{2} = \boxed{16}$$

11. From the figure below, $PB = 4AP$, $\frac{BQ}{QC} = \frac{2}{7}$, and the area of the quadrilateral $APQC$ is $37\,\text{cm}^2$. Find the area of triangle $BPQ$.

A) $10\,\text{cm}^2$ B) $8\,\text{cm}^2$ C) $4\,\text{cm}^2$ D) $12\,\text{cm}^2$

Assuming




$$\frac{\left(\frac{5}{4}z\right)(9y)}{2} - \frac{z(2y)}{2} = 37$$

$$\frac{45yz}{8} - yz = 37$$

$$45yz - 8yz = 37 \times 8$$

$$37yz = 37 \times 8$$

$$yz = \frac{37 \times 8}{37} = 8$$

$$\Delta BPQ = \frac{2yz}{2} = yz = \boxed{8}$$